User Manual

# FLIM Data Acquisition Card

**FL-DAQ-02**
**FL-DAQ-04**
**FL-DAQ-08**




User Manual – Version v1.2.4
12 Nov 2026

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# 1 General Information

This chapter provides essential information on the purpose of this manual, the manufacturer, regulatory compliance, warranty conditions, and the conventions used throughout this document.

## 1.1 About This Manual

This manual provides instructions for the installation, configuration, operation, maintenance, and disposal of the FLIM Data Acquisition Card. It forms an integral part of the product and must remain accessible to the personnel responsible for the device. If the device is transferred to a third party, all accompanying documentation must be delivered with it.

This manual contains proprietary information and may not be reproduced or distributed, in whole or in part, without prior written consent of the manufacturer.

Before performing any operation, the entire manual must be read carefully and the relevant sections reviewed in detail.

### 1.1.1 User Qualification

The device is intended for use by qualified personnel familiar with laboratory electronic instrumentation and responsible for the installation, operation, and maintenance of the device throughout its service life. It is not a toy and must not be used by children under 14 years of age. Persons with reduced physical, sensory, or cognitive abilities may operate the device only under supervision by qualified personnel.

### 1.1.2 Revision History

Document Revision: v1.2.4- 12 Nov 2026

A certified copy of this manual is stored in the manufacturer's technical file for the legally required period (10 years). During this period, documentation may be requested from the manufacturer.

## 1.2 Symbols and Conventions

The following graphical symbols and textual conventions may appear in this manual or on the device.

Provides important supplementary information.

Indicates procedures that, if not observed, may result in damage to the device or associated equipment.

Indicates procedures that, if not observed, may result in injury or serious damage.



**(a)** General hazard symbol



**(b)** Mandatory reading of the manual

**Figure 1.1:** The device may include the pictograms shown above.

All safety symbols must remain clearly visible throughout the lifetime of the device.

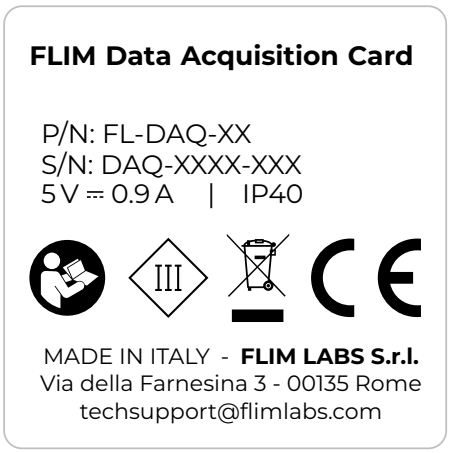



**Figure 1.2:** Product label

## 1.3 Manufacturer Information

| | |
|---|---|
| Manufacturer: | FLIM LABS S.r.l. |
| Address: | Via della Farnesina, 3 00135 Rome (RM), Italy |
| Telephone: | +39-329 761540 |
| E-mail: | info@flimlabs.com |
| Website: | www.flimlabs.com |

## 1.4 Product Label

The product label affixed to the device contains the required regulatory markings and identification information including:

- Product name and model
- Manufacturer identification
- Regulatory markings
- Electrical specifications

An example product label is shown in Figure 1.2. Disposal requirements associated with the WEEE marking are described in Section 7.2.

Safety pictograms performing a protective function must not be removed, covered, or damaged. If the label becomes illegible or damaged, contact authorized service for replacement.

## 1.5 Declaration of Conformity

The device is supplied with a Declaration of Conformity drawn up in accordance with applicable European legislation. Before use, verify that the Declaration of Conformity is present. A copy of the Declaration of Conformity is shown in Figure 1.3.

## 1.6 Warranty and Liability

### 1.6.1 Warranty

The manufacturer provides 24 months warranty for private customers and 12 months warranty for commercial customers. The warranty period begins on the date of delivery.

The warranty covers defects in materials and workmanship under normal and intended use. It is valid only if the device is operated and maintained according to this manual. Unauthorized modification, improper use, failure to follow instructions, or repair by non-authorized personnel voids the warranty. Full warranty terms are available in the manufacturer's General Terms and Conditions published on the official website.

To submit a warranty claim, provide the following:



- Product type/version
- Proof of purchase (including date)
- Detailed description of the issue

# EU Declaration of Conformity

CE

| Device Information | |
|---|---|
| Type: | Data Acquisition Card |
| Model: | FLIM Data Acquisition Card |
| Part Number: | FL-DAQ-02<br>FL-DAQ-04<br>FL-DAQ-08 |
| Serial Number: | As indicated on the product label |

| Manufacturer Information | |
|---|---|
| Manufacturer: | FLIM LABS S.r.l. |
| Address: | Via della Farnesina, 3 00135 ROME (RM), Italy |
| Telephone: | +39 – 329 761540 |
| E-mail: | info@flimlabs.com |
| Website: | www.flimlabs.com |

The present declaration of conformity is issued under the sole responsibility of the manufacturer. The device complies with the following directives and regulations:

**Applicable Directives and Regulations**

- Directive 2014/35/EU, known as the “Low Voltage Directive”
- Directive 2014/30/EU, known as the “Electromagnetic Compatibility Directive”
- Directive 2011/65/EU, known as “RoHS”
- Delegated Directives (EU) 2015/863 and (EU) 2017/2102 amending Directive 2011/65/EU
- Directive 2012/19/EU, known as “WEEE”
- Regulation (EU) 2023/988 on General Product Safety

The manufacturer undertakes to provide, upon a duly reasoned request from the national authorities, all relevant information regarding the device.

Signed for and on behalf of FLIM LABS S.r.l.:

**Place and Date**

Rome, 24/10/2025

**Signature**

Alessandro Rossetta, CEO

**Figure 1.3:** EU Declaration of Conformity

- Operation outside specified electrical or environmental limits
- Use of incompatible or non-compliant input signals or connected equipment
- Unauthorized modification, repair, or use of non-original spare parts
- Integration into systems not compliant with applicable safety regulations

The user is responsible for ensuring that the device is installed, operated, and integrated in accordance with this manual and all applicable local, national, and international regulations.

Except where otherwise required by mandatory law, the manufacturer shall not be liable for indirect, incidental, or consequential damages, including but not limited to loss of data, loss of profit, or system downtime.

# 2 Device Description

This section describes the functional characteristics, intended application, and technical features of the FLIM Data Acquisition Card.

## 2.1 Functional Description

The FLIM Data Acquisition Card (DAQ) is a compact data acquisition module designed for the acquisition, processing, and transmission of high-frequency digital electronic signals. It is intended for applications such as time-correlated measurements, event counting, and fluorescence lifetime imaging (FLIM) in scientific and laboratory instrumentation where high timing precision, reliability, and integration with external detection systems are required.

The device is designed to receive high-frequency digital input signals in LVTTL format (50 Ω) from external devices such as single-photon detectors, timing modules, or signal-conditioning electronics. Acquired signals are processed internally and transferred to a host system for analysis and data processing.

In addition to the detector input channels, the device provides dedicated synchronization inputs for Pixel Clock, Line Clock, and Frame Clock signals. These inputs allow the DAQ to correlate detected events with the position of a scanning system, enabling applications such as fluorescence lifetime imaging (FLIM), laser scanning microscopy, and other time-resolved imaging techniques.

The operating principle is based on programmable digital logic (FPGA), enabling flexible and deterministic processing of digital signals and precise management of timing events. The electronic architecture supports repetition frequencies up to 80 MHz, enabling high temporal resolution measurements. The device provides dedicated interfaces for synchronization, including high-frequency input trigger ports from external pulsed laser sources, trigger output ports for synchronizing external lasers or timing devices, and imaging synchronization inputs for pixel, line, and frame timing signals.

Communication with the host system is performed via USB, which is used both for data transfer and for device control and configuration. Power is supplied via 5 V DC directly from the USB interface.

The product is available in multiple variants that differ exclusively in the number of available input and output ports, while maintaining the same operating principle, technology, and safety characteristics.

The module features:

- FPGA-based digital processing for deterministic timing and event handling
- High-frequency detector inputs in LVTTL format (50 Ω) via SMA and LVDS via USB Type-C
- Dedicated Pixel Clock, Line Clock, and Frame Clock synchronization inputs for imaging applications
- Dedicated laser trigger input and laser trigger output for synchronization
- USB interface for configuration and data transfer (also providing 5 V power)
- Compact plastic enclosure
- Variant-dependent number of detector input channels (DAQ-02 / DAQ-04 / DAQ-08)

## 2.2 Intended Use

The FLIM Data Acquisition Card is intended as a compact electronic module for acquisition, processing, and transmission of electronic signals by professional users in private research laboratories, industrial environments, or universities. It may be used as a stand-alone unit or integrated into custom measurement systems and scientific instrumentation.

## 2.3 Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications
- Use with incompatible components
- Operation outside specified limits

- Unauthorized modification
- Use with incorrect power supply

## 2.4 Components Overview

Figure 2.1 shows an overview of the key components of the device. Those include:

1. USB micro-B port for power supply and data transmission to the host computer (PC)
2. Signal input ports available via SMA connectors and USB Type-C interfaces for connecting signals from detectors or external sources
3. Status LED indicators showing power status, operational status 1, operational status 2, and laser enable status
4. Reference signal ports available via SMA connectors and USB Type-C interfaces for connecting external timing signals
5. Laser synchronization input port (SMA or USB Type-C) for receiving the trigger signal from the laser source
6. Laser synchronization output port (SMA or USB Type-C) for distributing the trigger signal to external devices
7. Mounting holes on the enclosure for secure mechanical installation

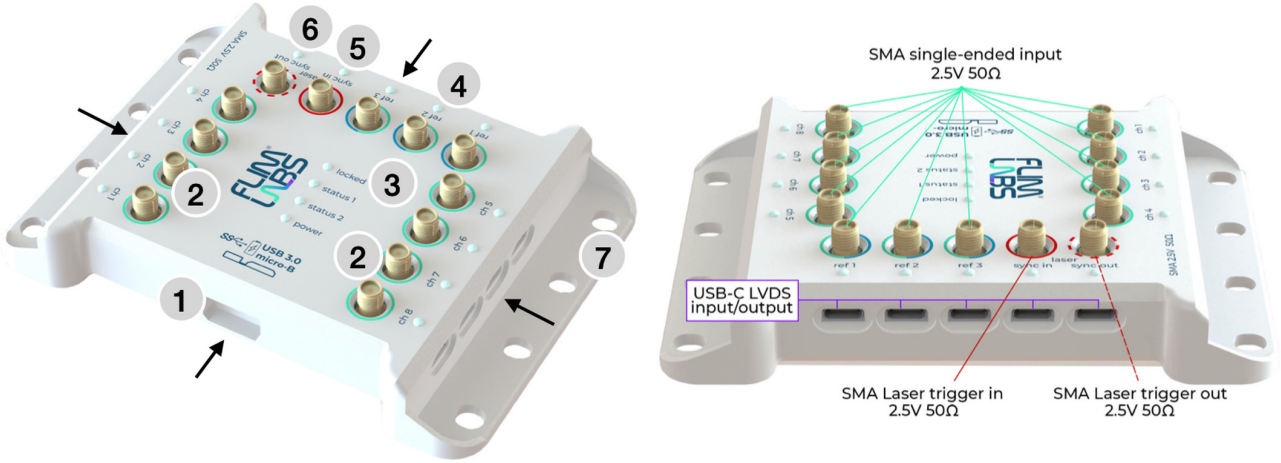



**Figure 2.1:** Overview showing the key components of the FLIM Data Acquisition Card.

## 2.5 Technical Specifications

The electrical, signal, and environmental limits referenced throughout this manual are summarized in this section.

| Item | DAQ-08 | DAQ-04 | DAQ-02 |
|---|---|---|---|
| **Electrical Specifications** | | | |
| Electrical safety class | Class III | | |
| Power supply | 5 V DC via USB micro-B | | |
| Maximum current consumption | 0.9 A | | |
| Maximum power consumption | 4.5 W | | |
| **Signal Specifications** | | | |
| Electronic hardware type | Field-Programmable Gate Array (FPGA) | | |
| Input signal type | SMA: 2.5 V digital LVTTL at 50 Ω<br>USB Type-C: LVDS | | |
| Input polarity | positive | | |
| Minimum input pulse width | > 1.5 ns | | |
| Minimum input pulse height (abs) | 2.5 V | | |
| Maximum input pulse height (abs) | 5 V | | |
| Single-ended SMA inputs (LVTTL) | 11 | 7 | 5 |
| USB Type-C LVDS inputs | 12 | 8 | 6 |
| Laser trigger input (SMA, sync input) | 1 | | |
| Laser trigger output (SMA, sync output) | 1 | | |
| Output signal standard | SMA: 2.5 V digital LVTTL at 50 Ω<br>USB Type-C: LVDS | | |
| Impedance | 50 Ω | | |
| PC connection | USB micro-B port | | |
| Timing variation (digital signal) | 80 ps RMS | | |
| Maximum repetition frequency | 80 MHz | | |
| Minimum resolvable fluorescence lifetime | 50 ps | | |
| Single-event precision ($\sigma/\sqrt{2}$) | 300 ps | | |
| Supported laser sync frequencies | Input: 10 MHz to 80 MHz<br>Output: 10, 20, 40, 80 MHz | | |
| Time bin width (temporal resolution) | 48 ps @ 80 MHz<br>96 ps @ 40 MHz<br>192 ps @ 20 MHz<br>384 ps @ 10 MHz | | |
| Dead time | 1.5 ns | | |
| Differential non-linearity | < 0.5% RMS | | |
| Maximum transfer rate (sum over channels) | 100 M counts/s | | |
| Maximum burst rate per input channel | 640 M counts/s | | |
| Minimum pixel dwell time | 100 ns | | |

| Item | DAQ-08 | DAQ-04 | DAQ-02 |
|---|---|---|---|
| Maximum duty cycle of reference signals | 80 % | | |
| Maximum acquisition duration | Not limited by the acquisition card | | |
| **Software Requirements** | | | |
| Acquisition software | FLIM Labs *Spectroscopy Py* or *FLIM IMAGER* | | |
| Operating system | Windows 10 or later | | |
| PC hardware | multicore CPU, 4GB RAM | | |
| **Physical Specifications** | | | |
| Dimensions (L x W x H) | 139 mm x 101 mm x 28 mm | | |
| Weight | 120 g | | |
| Protection rating | IP40 | | |
| **Environmental Specifications** | | | |
| Operating temperature | 0°C to +40°C | | |
| Storage temperature | -20°C to +70°C | | |
| Relative humidity | Up to 80% non-condensing | | |

**Table 2.2:** Technical specifications of the FLIM Data Acquisition Card.

## 2.6 Mechanical Drawings and Dimensions

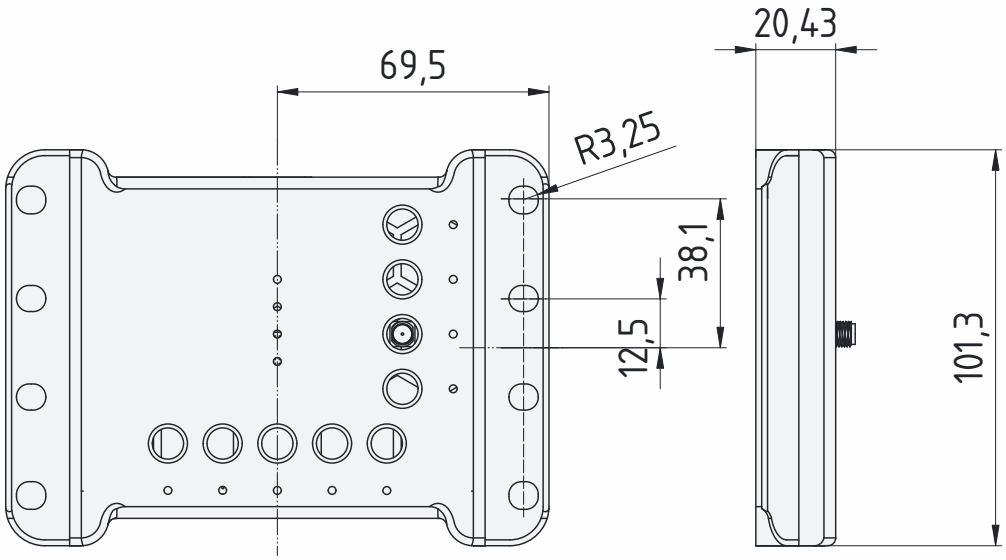



**Figure 2.2:** Technical drawing of the FLIM Data Acquisition Card.

# 3 Safety Information

## 3.1 General Safety Instructions

Before installation and operation, ensure that this manual is complete and legible. If any part of the documentation is missing or unclear, contact the manufacturer before proceeding.

The device must be used only as described in this manual. Any operation not explicitly described in this manual is considered improper use.

This manual complements applicable national and local safety regulations. If local regulations impose stricter requirements, those regulations take precedence.

Despite the implemented safety measures, residual risks may remain if the device is not used as specified.

## 3.2 Electrical Safety

- Operate the device only with the specified power supply.
- Use undamaged, properly rated cables.
- Disconnect power before modifying signal connections.
- Do not open the enclosure. There are no user-serviceable internal components.
- Keep away from water sources and other liquids.
- Unauthorized repair or modification is not permitted.

## 3.3 Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 0°Cto +40°C
- Storage temperature: −20°Cto +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases.

Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.

## 3.4 Modification and Improper Use

The device must not be modified, altered, or repaired by unauthorized personnel. Use outside the intended application, operation beyond specified electrical limits, or failure to follow the procedures described in this manual may result in malfunction and void the warranty.

## 3.5 Documentation Retention

This manual must be kept accessible to personnel responsible for installation, operation, and maintenance. A replacement copy may be requested from the manufacturer if required.

# 4 Setup

This section describes the correct procedure for unpacking, installing, and configuring the device before operation.

## 4.1 Scope of Delivery

- Data acquisition card (DAQ)
- USB 3.0 SuperSpeed micro-B cable
- Quick Start Guide
- Declaration of conformity
- Software for acquisition and data analysis is available for download from the manufacturer website under .

Verify that all items are present before proceeding with installation.

## 4.2 Unpacking and Inspection

The device is supplied in protective packaging to protect it during storage and transport. Handle the device carefully during unpacking to avoid mechanical damage.

After removing the packaging:

- Inspect the device and accessories for visible damage.
- Check for dents, deformation, cracks, or signs of impact.
- Verify that connectors and cables are intact.

If any damage, missing parts, or abnormalities are detected, stop the installation and contact the manufacturer before proceeding.

Dispose of packaging materials responsibly and keep them out of reach of children.

The device is compact and lightweight and may be moved manually. When handling, support it securely to prevent accidental dropping.

## 4.3 Installation Requirements

Before installation and operation, ensure that the environmental conditions specified in Section 3.3 are met. Children must not have access to the installation or storage area.



Before connecting external devices to the DAQ inputs, verify that the input signal characteristics comply with the requirements specified in Section 2.5. Signals outside the specified limits may damage the device or lead to unreliable operation.

## 4.4 Mechanical Installation

Place the device on a stable laboratory surface with adequate space for cable routing and ventilation.

Ensure that:

- The device is positioned to prevent mechanical stress on connectors.
- Cables are routed safely and do not create tripping hazards.
- The installation area is dry and away from water sources (e.g., sinks or splash zones).

Remove any protective dust or moisture covers before operation.

## 4.5 Electrical Connections

All electrical connections must be performed by qualified personnel who have read this manual. Connector locations are shown in Figure 2.1.

### USB Connection (Power and Data)

Connect the device to the host computer using the USB 3.0 micro-B cable. The USB connection provides both 5 V DC power supply to the device as well as data transfer and communication with the host PC. The following ports can be used

- USB Type-C using an adapter that supports USB 3.0 or higher.
- USB Type-A that supports USB 3.0, typically identified by a blue insert.



The device must be connected directly to the PC. Do not connect the device to a USB hub. USB 2.0 ports are not supported.

### Signal Input Connections

Connect external signals to the device using one of the supported interfaces:

- Single-ended SMA inputs (LVTTL, 50 Ω) using compatible SMA coaxial cables.
- USB Type-C LVDS inputs using compatible USB Type-C signal cables.

The number and type of available ports depend on the device model.



The device requires positive signals 2.5 V to 5 V with 50 Ω termination and triggers on the rising edge. Other signals may not be detected properly or could lead to device damage. Please check Table 2.2 for full signal requirements.

### Laser Trigger and Synchronization Connections

If required by the application (e.g., lifetime analysis), connect the laser synchronization signals as follows:

- Connect the laser trigger signal from the external laser source to the laser trigger input port.
- Connect the laser trigger output port to external devices requiring synchronization (e.g., laser modulation input or timing distribution).

If you are acquiring image data from a scanning system such as a microscope, you need to connect the corresponding trigger signals.

- Connect the pixel, line, and frame clock signals from your scanning system to the three reference signal ports of the DAQ.

See Figures 4.1 and 4.2 for examples of timing and image scanning configurations.

### Status Indicators

The device includes status LEDs indicating power state and operational states. During correct operation, verify that the LEDs indicate normal device status according to the intended configuration.

## 4.6 Software Installation

The FLIM Data Acquisition Card requires compatible software for configuration, data acquisition, and analysis. FLIM LABS provides dedicated software tools such as *Spectroscopy Py* and *FLIM IMAGER*.



This section provides only a basic overview. Please check the corresponding software manual available on the website for updated information, detailed installation instructions, configuration steps, and software-specific features.

### System Requirements

The following minimum requirements apply for running the software:

- PC with multicore CPU
- 4 GB RAM
- USB interface for connecting the DAQ to the PC
- Windows 10 or later

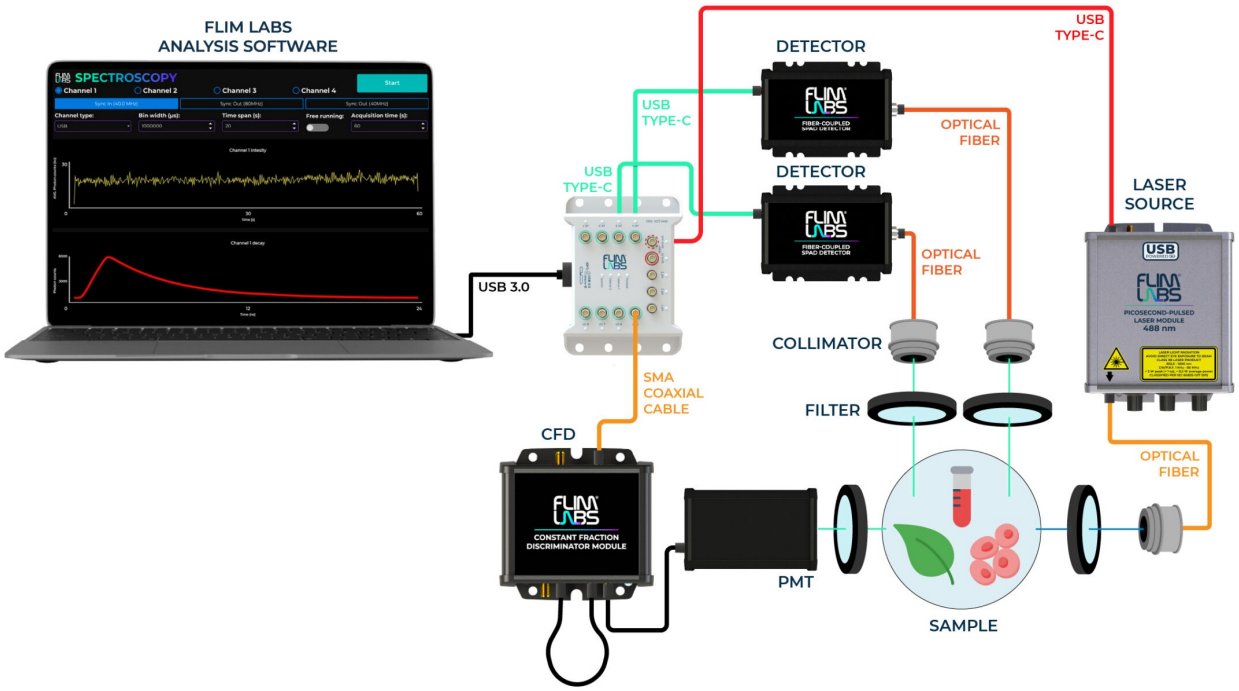



**Figure 4.1:** Example of a fluorescence lifetime analysis system using multiple detectors.

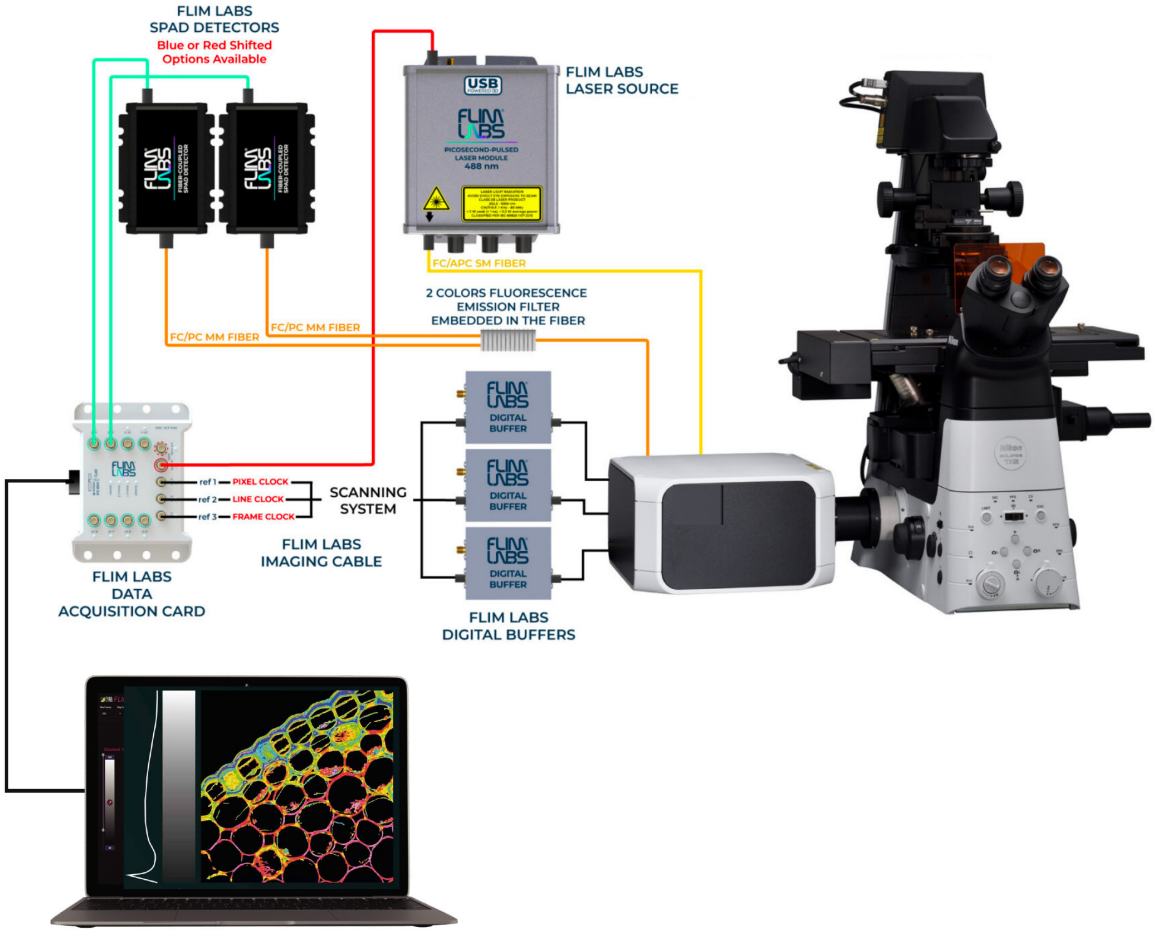



**Figure 4.2:** Example configuration of an image scanning setup.

Additional requirements may apply depending on the selected software (e.g., Python installation for advanced features).

### Installation

The software packages can be downloaded from the FLIM LABS website:

- Spectroscopy Py: https://flimlabs.com/spectroscopy-py/
- FLIM IMAGER: https://www.flimlabs.com/flim-imager/

1. Download the desired software package.
2. Run the installer or follow the setup instructions provided.
3. Connect the FLIM Data Acquisition Card to the PC. The drivers for correct hardware initialization are included in the installer package.

## 4.7 Initial Configuration

Functional configuration of the device is not performed through direct hardware commands, but exclusively through FLIM LABS software applications *Spectroscopy Py* or *FLIM IMAGER*. These applications provide the user interface for channel configuration, acquisition control, and visualization of the acquired data.

After completing all hardware connections and installation of the software, proceed with the following initial configuration steps:

1. Start the desired FLIM LABS software application (e.g., *Spectroscopy Py* or *FLIM IMAGER*).
2. If the device was connected before launching the software, the automatic device detection feature should identify the device. If the device is not detected automatically, press the *Check device* button to initiate the device search.
3. Once the device is detected, configure the acquisition channels. The software allows automatic channel identification using the *Detect channels* function, or manual selection of the desired input channels. Verify that the connection type corresponds to the physical interface used (USB Type-C or SMA).
4. Confirm that the input channels selected in the software correspond to the signals physically connected to the device. Only the channels selected in the software will be displayed and processed during acquisition.

Detailed procedures for software configuration, acquisition modes, data processing, and result visualization are described in the dedicated software documentation and are not part of this hardware manual.

If the device does not operate as expected after configuration, refer to the troubleshooting guidance provided in Section 5.5.

# 5 Operation

This section describes the functional behavior of the FLIM Data Acquisition Card during use, including its operating principle and normal operating conditions.



Do not operate the device if it is damaged or shows signs of unsafe condition.

## 5.1 Principle of Operation

The DAQ receives high-frequency digital input signals from external devices, processes them deterministically using FPGA-based logic, and transfers the acquired data to the host system via USB for analysis.

Depending on the software configuration and application mode, the device can:

- Count events on one or multiple input channels
- Perform timing-based measurements using external synchronization signals
- Support FLIM-related acquisition modes requiring laser trigger reference signals

The internal FPGA architecture ensures deterministic signal processing and precise timing of detected events, enabling high-resolution temporal measurements.

## 5.2 Software Control

The data acquisition card is controlled entirely through FLIM LABS software applications running on a host computer. The device can be operated using the software applications *Spectroscopy Py* or *FLIM IMAGER*. These applications provide the user interface for configuration, acquisition control, and visualization of the acquired data.

Typical software functions include:

- Selection and activation of acquisition channels
- Configuration of acquisition parameters
- Synchronization with external laser trigger signals
- Real-time visualization of photon counts and decay curves
- Export of acquired data for further analysis

Detailed instructions for software configuration, acquisition modes, data analysis, and visualization are provided in the dedicated software manuals available on the website under https://www.flimlabs.com/software/ and are not part of this hardware manual.

## 5.3 Normal Operation

After correct installation and configuration (see Chapter 4), the device operates under control of the host software. The host software configures the acquisition parameters, controls the start and stop of measurements, and displays the acquired data in real time.

Ensure that:

- The selected channels in the software match the physically connected inputs.
- Input signal levels and timing characteristics remain within the specified limits.
- The USB connection remains stable to avoid acquisition interruptions.

## 5.4 Parameter Adjustment

The DAQ does not provide user-adjustable hardware parameters for standard operation. Configuration of channels, acquisition modes, synchronization settings, and other acquisition parameters is performed through the dedicated FLIM LABS software.

If unexpected behavior is observed during operation, refer to Section 5.5.

## 5.5 Troubleshooting

Common faults and corrective actions are summarized in Table 5.1.

**Table 5.1:** Overview of common problems and recommended corrective actions.

| Problem | Possible Cause | Recommended Action |
|---|---|---|
| **Power and USB Communication** | | |
| Device does not power on | USB connection not providing power or cable defective | Use the supplied USB 3.0 micro-B cable. Try a different USB port and verify that the host provides sufficient power. |
| Device not detected by PC | USB data connection not established or software issue | Verify USB connection and restart the software. Try a different USB port or cable. Reinstall or update the FLIM LABS software if necessary. |
| Acquisition stops unexpectedly | USB communication interruption or host performance limitation | Ensure stable USB connection and avoid running resource-intensive applications during acquisition. |
| **Signal Input and Channel Configuration** | | |
| No data on one or more channels | Channel not enabled or mapped incorrectly in software | Verify that the selected channels correspond to the physically connected inputs. |
| Channels show zero counts | Detector or signal source not active | Verify that the detector or signal source is powered and producing valid digital pulses. |
| No data on one or more channels | Input signal does not meet LVTTL requirements | Verify input signal amplitude (2.5 V min, 5 V max), polarity, and pulse width (>1.5 ns). |
| Signal appears on wrong channel | Input cable connected to different physical channel | Verify physical channel numbering and software channel selection. |
| **Timing and Synchronization** | | |
| Incorrect timing results | Missing or incorrect synchronization signal | Verify laser trigger input connection and ensure correct sync source is selected in the software. |
| Unstable acquisition or missing events | Input frequency out of range or signal integrity issues | Verify repetition frequency (up to 80 MHz) and use proper 50 Ω coaxial cabling. |
| Distorted or irregular timing signals | Improper impedance matching or cable reflections | Use short 50 Ω cables and proper termination. Avoid T-connectors. |
| **Acquisition Performance** | | |
| Data rate lower than expected | USB bandwidth limitation or host performance constraint | Use a high-speed USB port and avoid USB hubs. Reduce active channels or acquisition settings if needed. |
| Acquisition cannot start | No acquisition channel enabled | Activate at least one acquisition channel in the software before starting acquisition. |



If any unexpected or potentially hazardous anomaly occurs during startup or operation, switch off the device immediately and disconnect it from the power source. Contact technical support before further use.

# 6 Maintenance and Service

This section describes the cleaning, preventive maintenance, corrective maintenance, and authorized service procedures for the device. All maintenance activities must be performed by qualified personnel who have read and understood this manual.



Before performing any cleaning or physical maintenance operation, the device must be switched off and disconnected from the power source.

Maintenance must be carried out in compliance with this manual and applicable workplace safety regulations. The maintenance area should provide adequate illumination; a minimum lighting level of 200 lux is recommended.

## 6.1 Cleaning

The device requires periodic external cleaning to ensure reliable operation.

- Remove dust or dirt from the exterior surfaces using a dry or slightly damp lint-free cloth.
- A mild, neutral detergent may be used if necessary.
- Do not allow liquids to enter connectors or internal components.
- Avoid the use of solvents, aggressive chemicals, or abrasive materials.

For connectors and hard-to-reach areas (e.g., USB ports), a soft-bristled brush or compressed air may be used.

## 6.2 Preventive Maintenance

The device does not require routine internal maintenance.
Periodically inspect:

- External surfaces for mechanical damage
- Connectors and cables for wear or deformation
- Product label for legibility. See Section 1.4 for label content and handling requirements.

If any abnormal condition is detected, discontinue use and contact technical support.

## 6.3 Service and Repairs

The device contains no user-serviceable internal components. In the event of malfunction, accidental damage, or improper operation:

- Switch off and disconnect the device.
- Do not attempt internal repair.
- Contact the manufacturer’s technical service for instructions.

Repairs or modifications must be carried out exclusively by FLIM LABS S.r.l. or authorized personnel. Use only original FLIM LABS spare parts. Further warranty conditions are described in Section 1.6.1.



Unauthorized intervention or the use of non-original components may compromise safety, performance, and warranty coverage.

# 7 Decommissioning and Disposal

This section describes procedures for removing the device from service and disposing of it in accordance with applicable regulations.

## 7.1 Decommissioning Procedure

The device is designed for long-term laboratory use. When it reaches the end of its technical or operational life, it must be decommissioned to prevent further use.

Decommissioning requires:

- Disconnecting the device from all power sources.
- Removing all signal connections.
- Ensuring that the device cannot be unintentionally reused.

The same procedure applies in cases of:

- Permanent withdrawal from service
- Storage without intended future use
- Final disposal

Reuse of components after final decommissioning is not permitted unless explicitly authorized by the manufacturer.

## 7.2 Disposal

The device must be disposed of in accordance with applicable local and European regulations governing electrical and electronic equipment. In accordance with Directive 2012/19/EU (WEEE), the device must not be disposed of with household waste. The product label includes the WEEE symbol indicating the requirement for separate collection of electrical and electronic equipment (see Section 1.4). Contact your local waste management authority or certified collection center for proper disposal procedures.

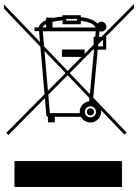


**Figure 7.1:** WEEE symbol